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A Regime-Switching Framework for Tactical Sector Tilts under Higher-Moment Investor Preferences

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Abstract

This thesis develops and evaluates a regime-aware core-plus-satellite allocation framework designed for implementation within the operational constraints of a wealth management context. A three-state Hidden Markov Model is estimated on monthly excess log returns of the S&P 500 Total Return index and the Bloomberg US Treasury 7–10 Year index. The resulting one-step-ahead predictive regime probabilities serve as the signal for a scoring engine that evaluates nine US sector ETFs and gold as candidate tilts against a 60/40 benchmark with a fixed 40% Treasury allocation and a replaceable 60% equity sleeve. Tilt scoring follows a Taylor-expanded expected utility function after Jondeau and Rockinger (2006), accommodating mean-variance, mean-variance-skewness, and mean-variance-kurtosis investor types. The central design feature is a rolling 60-month estimation window that enables the model to adapt to structural breaks in sector leadership, specifically the rotation from energy sector dominance during the 2003–2008 commodity supercycle to technology sector dominance in the post-2016 period. A score improvement floor functions as a hurdle rate, returning the portfolio to the passive 60/40 core when regime conviction is insufficient. The primary configuration (45% satellite sleeve, score improvement floor = 0.001) achieves a CAGR of 8.60–8.90% against a benchmark of 7.71%, Sharpe ratios of 0.920–0.956 versus 0.875, and maximum drawdown improvements of approximately 335 basis points. A conservative 20% sleeve variant is also reported, which simultaneously beats the benchmark on all five risk-adjusted metrics including downside deviation.

Keywords: Hidden Markov Model, regime switching, tactical asset allocation, sector rotation, higher-moment preferences, rolling window estimation, core-plus-satellite portfolio

JEL Classification: G11 (Portfolio Choice; Investment Decisions) · G17 (Financial Forecasting and Simulation) · C22 (Time-Series Models) · C61 (Optimization Techniques) · G23 (Pension Funds; Other Private Financial Institutions)

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1. Introduction

In the wealth management business, only a small percentage of clients are managed actively, and that depends largely on the size and the cost of continuing operations of the firm. Most individual clients and their portfolios are managed passively or on very infrequent rebalancing schedules. These portfolios are usually standardised for several reasons: to facilitate the communication chain between client, relationship manager, portfolio manager, and trading desk, and to abide by rules and regulations.

Once a standardised approach has been developed for a firm, an operational pipeline follows, with internal structures and processes built to maintain the order of business for smaller clients. In recent years these processes have been optimised across the industry, but their scale depends, again, on the size and cost of operations. According to PwC’s Asset and Wealth Management 2025 report, passive AUM is projected to reach USD 70 trillion by 2030 at an annualised rate of 10%, and 89% of asset managers reported profitability pressure over the five years preceding the report, with profit per AUM declining by approximately 19% since 2018. This is the operational economics argument that motivates the study: under cost pressure and growing AUM, the practical default for smaller clients is a standardised passive mandate.

This poses the question at the core of this study: is it possible to develop a model with a decision tilt that outperforms a passive benchmark on a risk-adjusted basis and that simultaneously addresses the concerns of investors with different preferences for higher moments? Most studies that use a regime switching framework to solve a portfolio allocation problem do so considering different types of investors, but not the type of portfolio manager who operates within a standardised, constrained mandate. This study deliberately caters to that audience. The final version of the model developed here is designed for a wealth management context, though it should not be excluded from other domains in the study of regime switching models, portfolio allocation signals, and sector risk-return analysis.

The main empirical results are reported in **Table 7**. We contribute to the regime-switching and portfolio allocation literature in several ways. First, our regime-switching model is integrated within a rolling 60-month estimation window that refits the HMM at every rebalance date, allowing the model to adapt to structural breaks in sector leadership. Second, we introduce scoring for allocation decisions rather than solving an optimisation problem, making the approach computationally manageable within a real operational pipeline. Third, we incorporate multiple investor preference types, from mean-variance to mean-variance-kurtosis, within the same framework, allowing the model to serve clients with different attitudes toward tail risk. Finally, we integrate a score improvement floor that functions as a hurdle rate, ensuring that the model only deviates from the passive core when the regime signal provides sufficient expected utility justification.

The academic literature on asset allocation under regime switching was opened by Ang and Bekaert (2002), who showed that the welfare cost of ignoring international diversification is significantly amplified in bear regimes, particularly when a risk-free asset is available. Ang and Bekaert (2004) subsequently demonstrated in a practitioner-oriented paper that regime-based allocation strategies outperform static 60/40 even after transaction costs, with the key insight being that the benefit comes from regime persistence rather than prediction. Guidolin and Timmermann (2007, 2008) extended this framework to a four-state model and showed that optimal allocations differ substantially across regimes and that ignoring regime switching generates welfare costs of several cents per dollar of initial wealth at short investment horizons, rising above eight cents at 60-month horizons under high risk aversion. The present study sits in this tradition but departs from it in three important ways. We use a simpler, operationally manageable scoring approach rather than full Bellman equation optimisation. We focus on a US sector ETF universe rather than international equity. And we introduce rolling window re-estimation as the mechanism for handling structural breaks, an aspect that was not required in studies relying on the stable 1954–1999 data environment.

The thesis is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the data. Section 4 presents the Hidden Markov Model specification and estimation. Section 5 develops the investor preference theory and scoring function. Section 6 describes the allocation layer architecture. Section 7 presents the satellite tilts, conviction scalar, and rolling window mechanics. Section 8 reports the key findings. Section 9 discusses gold as a conditional safe haven. Section 10 concludes.

2. Literature Review

2.1 Regime Switching in Asset Allocation

The regime-switching framework for financial time series originates with Hamilton (1989), who introduced the Markov-switching model as a means of capturing structural change in macroeconomic time series. Hamilton's key contribution was the filtered probability recursion, which allows the latent state to be inferred from observed data using the EM algorithm, producing a probabilistic assessment of the current regime rather than a binary classification. This framework was subsequently extended to financial applications by a growing literature documenting the presence of regimes in both stock and bond returns.

The first direct application to portfolio allocation under regime switching is Ang and Bekaert (2002), who evaluate international asset allocation in the presence of bear and bull regimes. Using a two-state HMM on international equity returns, they show that correlations increase substantially during bear regimes, eroding the diversification benefit that static mean-variance analysis predicts. The welfare cost of ignoring these regime-dependent correlations is estimated at two to three cents per dollar of initial wealth when a conditionally risk-free asset is included in the portfolio; in the all-equity case the cost is substantially lower, at around 0.2 to 0.4 cents. They find that optimal portfolio weights differ significantly across regimes. Ang and Bekaert (2004) extend this to a practitioner-oriented study, demonstrating that regime-aware allocation strategies outperform a static 60/40 benchmark even after transaction costs, with the key mechanism being regime persistence: the model generates value because regimes last long enough for the allocation signal to be actionable.

Guidolin and Timmermann (2007) provide the most comprehensive treatment of multivariate regime switching for asset allocation, estimating a four-state model on US stocks and bonds over 1954–1999 and solving the full dynamic programming problem for buy-and-hold and rebalancing investors with power utility. Their four states, characterised as crash, slow growth, bull, and recovery, exhibit very different means, volatilities, and cross-asset correlations, and optimal allocations vary substantially across states. The welfare cost of ignoring regimes is reported in cents per dollar of initial wealth (Table 5 of that paper), reaching approximately eight cents at a 60-month horizon under high risk aversion. The paper documents out-of-sample performance evaluation across multiple investor types and horizons and finds that the regime model outperforms simpler benchmarks across most investment horizons. A key finding relevant to the present study is that no VAR terms are required in the model, suggesting that the return distribution itself, rather than predictive variables, carries the relevant information. Guidolin and Timmermann (2008) extend this to international equity markets, incorporating four-moment preferences and showing that regime switching generates endogenous non-normality in portfolio returns, with co-skewness and co-kurtosis properties playing an important role in home bias. The Ang and Timmermann (2012) sur-

vey subsequently documents that multi-state specifications, generally three or more states, outperform two-state models across a range of asset allocation applications, supporting the three-state specification used in the present study.

The present study contributes to this literature in three specific ways. First, it applies the HMM framework to the US sector ETF universe — a dimension of the allocation problem that the international equity focus of the preceding papers does not address. Second, it replaces the full Bellman equation optimisation with a operationally manageable scoring approach, making the framework viable within a wealth management pipeline. Third, and most importantly, it introduces a rolling 60-month estimation window as the mechanism for adapting to structural breaks in sector leadership, explicitly addressing the stale-moment problem that arises when an expanding window is used across multi-decade samples.

2.2 Sector ETF Allocation under Regime Switching

The application of regime switching to sector-level allocation is a relatively recent and narrower strand of the literature. Ma, MacLean, Xu, and Zhao (2011) is the closest antecedent to the present study, proposing a regime-dependent factor-risk-neutral strategy for the nine State Street SPDR sector ETFs, the same universe used here. They model factor-risk-neutral allocation using a Markov regime-switching framework and compare performance against an equal-weight benchmark across the same ETF universe. Their results demonstrate that regime-dependent sector allocation outperforms the static benchmark, and they use the same sector ETF selection criteria applied here: the nine ETFs are from a single provider, have a long trading history beginning December 1998, and provide comprehensive coverage of the US market. The key difference between Ma et al. (2011) and the present study is that their framework does not include a fixed bond core, does not incorporate higher-moment investor preferences, and uses an expanding estimation window. The rolling window adaptation and the core-plus-satellite architecture are specific contributions of the present study.

Dou, Gallagher, Schneider, and Walter (2014) provide direct empirical evidence that cross-sector regime-dependent allocation outperforms cross-regional allocation on Sharpe ratio, return, and beta in both in-sample and out-of-sample periods over 1995–2010. Their unconstrained regime-switching sector strategy achieves an out-of-sample Sharpe ratio of 0.93 compared to 0.30 for the world market portfolio, with notably lower beta in bear regimes. Under a benchmark constraint, the out-of-sample Sharpe narrows to 0.39 versus 0.32. Their finding that sector allocation is more amenable to regime-based improvement than regional allocation is consistent with the sector ETF focus of the present study, and their documentation of lower cross-sector correlation during bear regimes supports the theoretical motivation for the conviction scalar used in the allocation layer here. Jiang, Liu, and Tse (2015) demonstrate similar results using international iShares ETFs with a daily rebalancing MRS strategy, finding outperformance after transaction costs and providing additional evidence that ETF-

based regime switching strategies are practically implementable at the institutional level.

2.3 Higher-Moment Preferences and Taylor-Expanded Utility

The theoretical foundation for incorporating skewness and kurtosis into investor preferences is provided by Scott and Horvath (1980), who use proof by contradiction under the assumption of decreasing absolute risk aversion to establish the sign of investor preference for moments of higher order than variance. Their result that the third moment of the utility function is positive (investors prefer positive skewness) and the fourth moment is negative (investors dislike excess kurtosis) holds broadly for risk-averse utility functions, not just specific parametric forms. This generality is important: it means the investor types defined in this study do not depend on a specific assumed utility function, but on a universal property of risk-averse behaviour.

Jondeau and Rockinger (2006) provide the operational implementation of these preferences through a Taylor series expansion of expected utility around the moments of the return distribution. Their four-moment approximation gives a closed-form scoring function that is accurate for moderately non-normal distributions and that handles a large number of assets without requiring the computationally prohibitive co-skewness and co-kurtosis tensors that direct four-moment optimisation would require. A key finding in Jondeau and Rockinger (2006) is that the opportunity cost of using only mean and variance to approximate expected utility ranges from negligible in near-normal data environments to more than 25 cents per dollar in highly non-normal settings such as emerging market or individual stock return distributions, confirming that higher moments carry economically meaningful information when return distributions are sufficiently non-normal. They also present and evaluate a full four-moment MVSK investor who combines skewness preference and kurtosis aversion in a single objective function. The present study deliberately separates these into distinct MVS and MVK investor types, allowing the contribution of each moment to be isolated empirically. Kim, Fabozzi, Cheridito, and Fox (2014) demonstrate that robust mean-variance portfolios implicitly favour skewness and penalise kurtosis without requiring explicit higher-moment terms, providing a complementary argument for the tractability of this approach.

2.4 Rolling Window Estimation and Time-Varying Parameters

The rolling window estimation approach used in this study is grounded in the parameter instability literature. Pesaran and Timmermann (2007) establish that in the presence of structural breaks, the optimal estimation window trades off bias from stale data against variance from insufficient observations, and that rolling windows are preferred over expanding windows when the data-generating process exhibits parameter instability. Their theoretical results provide the formal justification for preferring a fixed-length rolling window over the expanding window used in the earlier regime-switching literature. The Hematizadeh, Tajdini, and Hallahan (2022) time-varying transition probability study is also relevant: their

finding that even constant-transition-probability HMMs with rolling window estimation capture most of the time-variation benefit of more complex TVTP models provides support for the simpler constant- P specification used here. Nystrup, Hansen, Madsen, and Lindström (2015) demonstrate that regime-based allocation with adaptive parameter estimation outperforms static allocation without requiring return forecasting skill, with the key mechanism being regime persistence rather than prediction accuracy. Their use of a confidence threshold before changing positions is the conceptual antecedent of the score improvement floor mechanism in the present study.

The rolling window specification in this study introduces a specific design tension worth documenting in the literature. The number of states, $k = 3$, is selected on the basis of the full 27-year sample. The rolling 60-month estimation then uses this established structural assumption to re-estimate only the parameter values within each window. This two-stage design is analogous to model selection followed by rolling out-of-sample evaluation: the structural form is validated on the full sample, and the rolling window then adapts the parameters of that structure to the current data environment. The alternative of re-selecting k at every window would be computationally expensive, methodologically inconsistent, and likely to produce spurious variation in the number of states that would interfere with the allocation signal. The two-stage approach is both principled and practical, with one caveat: because $k = 3$ is selected by inspecting the full sample that includes the evaluation period, this is a modelling assumption rather than a fully out-of-sample structural choice. The rolling backtest tests the performance of this fixed structural assumption under recursively updated parameters.

3. Data

In order to define the regimes, one broad equity index and one broad bond index were used: the S&P 500 total return index and the Bloomberg US Treasury 7–10 Year index (LT09TRUU). These monthly returns were continuously compounded. The S&P 500 total return series was obtained through the Yahoo Finance API, and the Bloomberg Treasury index through Bloomberg. The 1-month T-Bill rate was subtracted from both to calculate the excess log return, which is how both series are referred to in this study. The monthly T-Bill simple return was acquired through the official Fama-French website and converted to log return for comparability with the other data. The start of the sample period for constructing the HMM is January 1999 and the end date is March 2026, matched to the issuance dates of the sector ETFs used for satellite tilts.

For the satellite universe, this study uses the nine State Street SPDR sector ETFs: Materials (XLB), Energy (XLE), Financials (XLF), Industrials (XLI), Technology (XLK), Consumer Staples (XLP), Utilities (XLU), Health Care (XLV), and Consumer Discretionary (XLY). Gold (XAU) is included as a tenth satellite candidate. All nine sector ETF return series were obtained through the Yahoo Finance API. The gold (XAU) return series was obtained from Bloomberg.

The selection of LT09TRUU as the bond component of the regime identification model was made after comparing three bond index variants: the Bloomberg US Aggregate Bond Index (LBUSTRUU), the Bloomberg US Treasury 7–10 Year index (LT09TRUU), and a mutual fund proxy tracking the Aggregate index (WFBIX). The Treasury-only index produced the most economically interpretable and stable regimes, particularly across the 2001 dotcom crash, the 2008 Global Financial Crisis, and the 2020 COVID-19 period. This is consistent with the intuition that a pure-rate-risk instrument with no credit exposure provides a cleaner signal for the bond-equity correlation dynamics that drive regime identification.

During the exploration phase of this project, monthly data from the EURUSD exchange rate, WTI crude oil, and bond indices at alternative durations were also tested with the Hidden Markov Model. Those outputs were either too noisy or could not be realistically interpreted in a macroeconomic context. Given the focus of the study on sector ETF tilts, their inclusion could not be reasonably justified at this stage of the research.

Figure 1 shows the full price history of all twelve assets. Together, the S&P 500 TR and LT09TRUU constitute the 60/40 benchmark core at a fixed weight allocation. The choice of a 60/40 split is the industry standard for a balanced client mandate and represents the passive baseline against which all strategy configurations are evaluated.

Rebalancing timeline and data construction. All series are monthly end-of-month log returns from total-return adjusted prices. Bloomberg indices are total-return series. The

no-look-ahead convention is strict: the HMM is estimated using returns through month t ; predictive probabilities $\pi_{t+1|t}$ are computed from the filtered state at t ; weights are set at the close of month t and earn returns in month $t + 1$. Transaction costs of 5 basis points per side are deducted at rebalancing. The risk-free rate is aligned to the same month-end.

Figure 1 — Cumulative Price Levels of Core and Satellite Assets (Feb 2004 = 100)

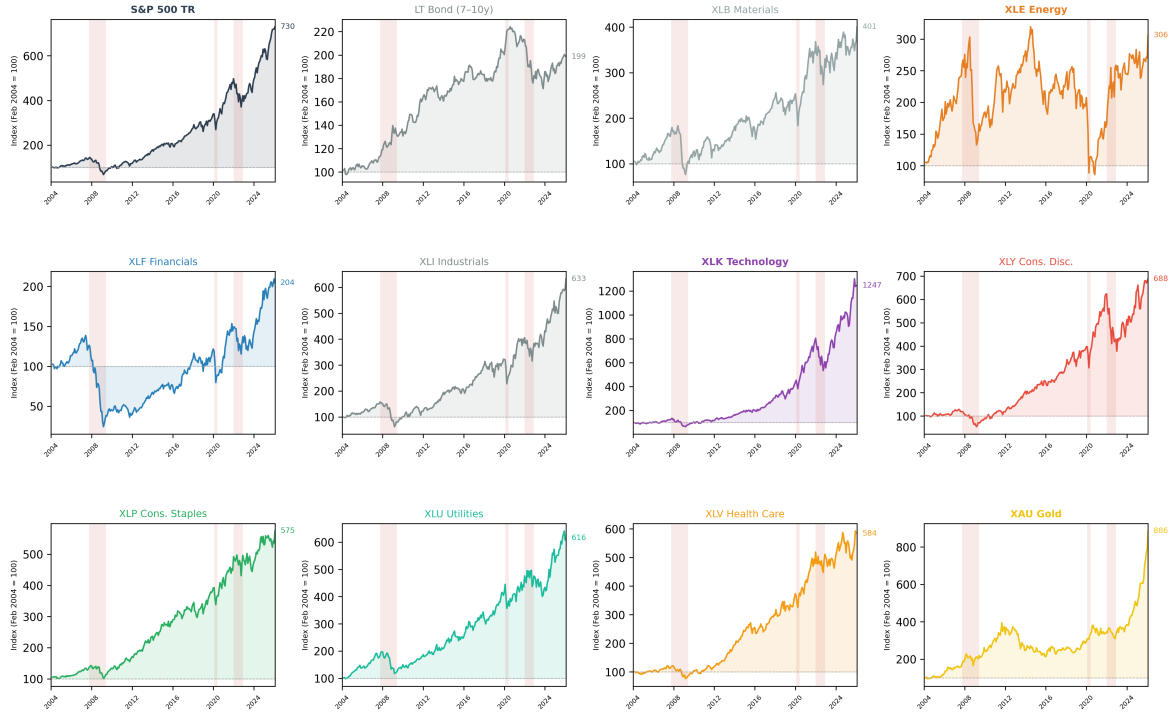


Figure 1: Cumulative price levels of the core and satellite assets, rebased to 100 at February 2004. Shaded regions mark the four crisis episodes. Final index values are annotated at the right of each panel. XLK (Technology, 1247) and XAU (Gold, 896) are the highest-returning satellites over the full period; XLE (Energy, 306) and XLF (Financials, 204) are the weakest cyclical performers, reflecting the post-2014 energy bear market and the persistent post-GFC underperformance of financials respectively.

Table 1: Descriptive Statistics for Monthly Log Returns — Full Sample (Feb 2004 – Jan 2026)

Asset	Ann. Mean (%)	Ann. Vol (%)	Sharpe	Skewness	Exc. Kurt.	Min (%)	Max (%)
<i>Core assets</i>							
S&P 500 TR	10.16	14.65	0.523	−0.806	1.831	−18.39	12.06
LT Bond (7–10y)	3.33	6.41	0.129	0.056	0.868	−4.80	7.69
<i>Cyclical sector satellites</i>							
XLB Materials	8.32	19.77	0.294	−0.567	1.842	−25.35	15.99
XLE Energy	8.63	25.99	0.236	−0.667	4.539	−42.12	26.82
XLF Financials	5.72	21.67	0.149	−1.120	4.116	−30.38	19.72
XLI Industrials	10.12	18.30	0.417	−0.733	2.334	−20.61	16.61
XLK Technology	13.14	17.92	0.594	−0.485	0.376	−17.58	12.87
XLY Cons. Disc.	10.52	18.53	0.433	−0.257	1.415	−19.40	17.30
<i>Defensive sector satellites</i>							
XLP Cons. Staples	8.66	11.69	0.527	−0.578	1.168	−13.48	9.94
XLU Utilities	9.31	14.21	0.479	−0.848	1.020	−13.93	10.03
XLV Health Care	8.98	13.62	0.476	−0.379	0.449	−13.23	11.86
<i>Commodity satellite</i>							
XAU Gold	11.36	16.75	0.529	−0.184	0.621	−18.50	12.49

Notes: Monthly log returns from the allocation backtest period (February 2004 – January 2026, $T = 264$ monthly observations). Annual mean and volatility obtained by multiplying monthly mean by 12 and monthly standard deviation by $\sqrt{12}$. Sharpe ratio uses an approximate average 1-month T-bill rate of 2.5% per annum as the risk-free rate. Skewness and excess kurtosis use Fisher’s definition. All nine sector ETF returns obtained via Yahoo Finance API; XAU obtained from Bloomberg; LT Bond = LT09TRUU Bloomberg US Treasury 7–10 Year index.

Table 2: Pairwise Correlation Matrix of Monthly Log Returns (Feb 2004 – Jan 2026)

	SP500	Bond	XLB	XLE	XLF	XLI	XLK	XLY	XLP	XLU	XLV	XAU
SP500	1.00	−0.06	0.86	0.63	0.85	0.92	0.90	0.90	0.72	0.51	0.75	0.09
Bond		1.00	−0.11	−0.25	−0.16	−0.11	−0.03	−0.03	0.10	0.22	0.04	0.32
XLB			1.00	0.68	0.75	0.87	0.71	0.77	0.63	0.43	0.65	0.24
XLE				1.00	0.55	0.63	0.46	0.48	0.42	0.33	0.43	0.13
XLF					1.00	0.84	0.63	0.75	0.59	0.34	0.62	−0.03
XLI						1.00	0.76	0.83	0.69	0.45	0.66	0.06
XLK							1.00	0.83	0.55	0.40	0.58	0.04
XLY								1.00	0.59	0.36	0.61	0.03
XLP									1.00	0.60	0.68	0.13
XLU										1.00	0.46	0.23
XLV											1.00	0.10
XAU												1.00

Notes: Pearson correlations of monthly log returns. Several features of this matrix motivate the satellite selection logic. XAU (gold) has near-zero correlation with all sector ETFs and a mildly negative correlation with bonds, confirming its role as a diversifier orthogonal to both core assets. XLE (energy) has the lowest correlation with the S&P 500 among cyclical sectors (0.63), explaining its frequent selection during the 2004–2009 commodity supercycle. XLU (utilities) has the lowest sector-to-S&P 500 correlation (0.51) among the cyclicals, consistent with its defensive classification. Bond–equity correlation is −0.06 unconditionally but varies substantially across regimes, as documented in **Table 4**.

4. Model

There are several criteria that predate the use of a HMM and are specific to this study. The outputs of the model need to have an intuitive economic interpretation that can be communicated to a portfolio manager, where we assume the portfolio manager is familiar with the concepts of expected return, standard deviation, and correlation, and understands their interpretation in a portfolio context. The identified regimes need to be stable across seeds and sample periods in order to reliably recognise key global events such as the dotcom crash, the Global Financial Crisis, and the COVID-19 pandemic. The outputs need to serve as input signals to the allocation layer. And the covariance between assets needs to be conditional on the regime, in order to capture the co-movement dynamics that inform the satellite tilt selection.

An important design point concerns what variables are used to define the regimes. This model is entirely endogenous: the regimes are identified exclusively from the joint distribution of the two return series, with no dividend yield, short-term interest rates, or macroeconomic variables introduced as exogenous predictors. This stands in contrast to a number of papers in the regime-switching literature, including the VAR-extended specification of Guidolin and Timmermann (2007) and the macro-conditioned models in the time-varying transition probability literature, which introduce external conditioning variables to capture predictability beyond the return distribution itself. The choice of an endogenous specification here is deliberate: it keeps the model self-contained within the information available from traded prices alone, which is consistent with the operational context of a wealth management firm that may not have reliable real-time macro feeds at the frequency and latency required for monthly rebalancing.

Regime-switching models for financial time series belong to a broader family of models that include threshold autoregressive models, hidden Markov models, hidden semi-Markov models, and smooth-transition models. Within this family, the Hidden Markov Model with Gaussian emissions was selected because it satisfies all four criteria above and because its filtered probability output is directly usable as a forward-looking allocation signal without additional transformation. GARCH-family models are widely used in the literature on co-movement and conditional heteroskedasticity, and this family of models produces valuable insights into the time-varying second moment structure of returns. However, GARCH outputs do not naturally produce discrete regime labels that partition the joint return distribution into economically interpretable states, which is a specific requirement of the allocation layer in this study. The HMM's filtered probabilities, by contrast, provide a direct probability distribution over a small number of interpretable states at each point in time.

4.1 HMM Specification

Let $\mathbf{Y}_t = (y_{1t}, y_{2t})'$ denote the bivariate vector of monthly excess log returns on the S&P 500 Total Return index and LT09TRUU at time $t = 1, \dots, T$. The Hidden Markov Model assumes that the conditional distribution of $\mathbf{Y}_t$ is governed by an unobservable discrete state variable S_t that takes values in $\{0, 1, \dots, K-1\}$ and follows a first-order Markov chain with constant transition probability matrix $\mathbf{P}$, where element $p_{ij} = \Pr(S_{t+1} = j \mid S_t = i)$.

Conditional on the state, returns are assumed to follow a multivariate normal distribution:

$$\mathbf{Y}_t \mid S_t = k \sim \mathcal{N}(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (1)$$

where $\boldsymbol{\mu}_k$ is the (2×1) vector of state-conditional mean excess returns and $\boldsymbol{\Sigma}_k$ is the (2×2) full covariance matrix for state k . The full covariance specification is essential: it allows the model to capture both state-specific volatility and state-specific cross-asset correlations, which are the mechanism through which the model captures regime-dependent bond-equity co-movement. As shown in **Table 4**, this correlation varies substantially across regimes and does not follow a simple negative-in-bear pattern.

The initial state distribution is parameterised as $\boldsymbol{\pi}_0 = (\pi_{0,0}, \dots, \pi_{0,K-1})'$ and estimated alongside the other parameters. The full parameter vector is $\boldsymbol{\theta} = \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k, \mathbf{P}, \boldsymbol{\pi}_0\}_{k=0}^{K-1}$.

4.2 Estimation

The model is estimated by maximum likelihood via the Expectation-Maximisation algorithm introduced by Hamilton (1989). The implementation uses the `hmmlearn` Python library, with custom pre-processing for excess log return construction and post-processing for regime ordering and rolling window management. The EM algorithm iterates between two steps. The E-step computes the filtered state probabilities using the Hamilton filter:

$$\xi_{t|t}(k) = \Pr(S_t = k \mid \mathbf{Y}_1, \dots, \mathbf{Y}_t; \boldsymbol{\theta}) \quad (2)$$

These filtered probabilities are the forward-pass output of the filter: they represent the probability of being in regime k given all information available up to and including time t . They are computed recursively from the prediction step:

$$\xi_{t|t-1}(k) = \sum_j \xi_{t-1|t-1}(j) p_{jk} \quad (3)$$

updated with the likelihood of the current observation in each state.

The M-step updates $\boldsymbol{\theta}$ by maximising the expected complete-data log-likelihood with respect to the smoothed state probabilities. Convergence is declared when the improvement in log-likelihood between iterations falls below 10^{-6} . To mitigate the risk of converging to local

optima, 25 random initialisations were run and the solution with the highest converged log-likelihood was retained.

Likelihood computation via the forward algorithm. The observed-data log-likelihood is obtained by integrating out the latent state sequence. For a single observation $\mathbf{Y}_t$ in state k , the state-conditional emission density is:

$$f(\mathbf{Y}_t | S_t = k; \theta) = \frac{1}{(2\pi)^{n/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{Y}_t - \mu_k)' \Sigma_k^{-1} (\mathbf{Y}_t - \mu_k)\right) \quad (4)$$

where $n = 2$ is the number of observed series. The full observed-data log-likelihood $\mathcal{L}(\theta) = \log p(\mathbf{Y}_1, \dots, \mathbf{Y}_T; \theta)$ is evaluated using the forward (filter) recursion. Define the forward variable:

$$\alpha_t(k) = p(\mathbf{Y}_1, \dots, \mathbf{Y}_t, S_t = k; \theta) \quad (5)$$

The recursion is initialised at $t = 1$ as:

$$\alpha_1(k) = \pi_{0,k} \cdot f(\mathbf{Y}_1 | S_1 = k; \theta) \quad (6)$$

and propagated forward for $t = 2, \dots, T$:

$$\alpha_t(k) = f(\mathbf{Y}_t | S_t = k; \theta) \sum_{j=0}^{K-1} \alpha_{t-1}(j) p_{jk} \quad (7)$$

The log-likelihood is then:

$$\mathcal{L}(\theta) = \sum_{t=1}^T \log \sum_{k=0}^{K-1} \alpha_t(k) \quad (8)$$

In practice, the recursion is implemented in log-space using the log-sum-exp trick to avoid numerical underflow, which is the approach taken in `hmmlearn`. The filtered probabilities from Equation (2) are recovered by normalising: $\xi_{t|t}(k) = \alpha_t(k) / \sum_j \alpha_t(j)$.

The following code illustrates the core estimation call. `GaussianHMM` from `hmmlearn` is used with `covariance_type="full"` to allow a distinct full covariance matrix per regime. Twenty-five random initialisations are run and the best log-likelihood solution is retained.

```
from hmmlearn import hmm
import numpy as np

best_model, best_ll = None, -np.inf

for seed in range(1, 26):
    model = hmm.GaussianHMM(
        n_components=3,
```

```

        covariance_type="full",
        n_iter=2000,
        tol=1e-6,
        random_state=seed,
    )
    model.fit(X)          # X: (T, 2) array of excess log returns
    ll = model.score(X)
    if ll > best_ll:
        best_ll, best_model = ll, model

# One-step-ahead predictive probabilities
filtered    = best_model.predict_proba(X)
predictive  = filtered @ best_model.transmat_

```

Post-estimation, regimes are reordered by ascending mean equity return so that Regime 0 is always the bear state, consistently across all seeds and rolling windows.

4.3 Model Selection and the Two-Stage Design Rationale

Three state configurations were evaluated: $K = 2$, $K = 3$, and $K = 5$. The two-state model collapsed the bear and transitional periods into a single high-volatility state, losing the distinction between genuine market stress and regime uncertainty. The five-state model produced average regime durations of one to two months with no stable economic interpretation, consistent with the overfitting behaviour documented in Guidolin and Timmermann (2007). The three-state model, $K = 3$, produced economically interpretable and statistically stable regimes across the full sample.

It is important to explain the two-stage nature of this design, as a potential methodological concern arises from the fact that $k = 3$ was selected by examining the full 27-year sample, while the actual allocation strategy uses a rolling 60-month estimation window. These are two separate and complementary uses of the model. The full-sample estimation is a diagnostic exercise: its purpose is to establish that three states are the appropriate structural assumption for this asset pair, that these states are stable enough to carry economic interpretation, and that they recur across multiple market cycles in a consistent way. This is the model selection stage. Once $k = 3$ is established as the correct structural assumption, the rolling window then takes this as given and re-estimates only the parameter values, not the number of states, within each 60-month window.

Models where any regime's expected duration was less than three to four months, or where any regime accounted for less than 6% of total observations, were considered too noisy and

Table 3: Model Selection: Log-Likelihood, AIC and BIC for $K = 2, 3, 5$ States

States	Log-lik.	Params	AIC	BIC	Avg. dur. (mo)	Min. regime frac.
$K = 2$	1427.86	13	-2829.72	-2780.53	11.0	0.234
$K = 3$	1445.83	23	-2845.67	-2758.64	15.3	0.062
$K = 5$	1466.68	49	-2835.35	-2649.94	4.7	0.009

Notes: Best log-likelihood across 25 random seeds for each K . Free parameters: $(K^2 - K)$ transition probabilities $+K \times 2$ means $+K \times 3$ unique covariance entries $+(K - 1)$ initial state probabilities. $AIC = -2\mathcal{L} + 2p$; $BIC = -2\mathcal{L} + p \ln T$, $T = 325$. $K = 3$ is selected: it has the best AIC (-2845.67) and adequate minimum regime fraction (6.2%). $K = 2$ has the best BIC but collapses the bear and transitional states into one, losing economic interpretability. $K = 5$ achieves the highest log-likelihood but produces a degenerate regime with only 0.9% of observations and an average duration of 4.7 months, consistent with overfitting. Bold = selected specification.

not developed further. The expected duration of regime k is:

$$\mathbb{E}[\text{duration}_k] = \frac{1}{1 - p_{kk}} \quad (9)$$

where p_{kk} is the diagonal element of the transition matrix for regime k . The three identified regimes are ordered by ascending mean excess equity return: Regime 0 is the bear state, Regime 1 is a transitional state, and Regime 2 is the bull state. This ordering is enforced consistently across all seeds and rolling windows.

Figure 2 shows the one-step-ahead predictive regime probabilities over the full 2004–2026 backtest period. The bear regime (Regime 0, red) spikes sharply during the GFC, COVID-19, and the 2022 rate shock, consistent with the rolling window adapting toward elevated bear-regime probability during these episodes. The transitional regime (Regime 1, orange) dominates the 2010–2016 recovery period, consistent with the sub-period alpha analysis in Section 8.

4.4 Model Output and Economic Interpretation

Table 4 reports the full regime-conditional moment structure, and **Table 5** presents the estimated transition probability matrix. The primary output of the HMM for allocation purposes is the one-step-ahead predictive probability vector:

$$\pi_{t+1|t} = \xi_{t|t} \cdot \mathbf{P} \quad (10)$$

This forward-looking vector encodes the model’s current belief about which regime will prevail in the next period. It is constructed using only information available at time t , ensuring no look-ahead bias in the backtest.

Additional outputs extracted from the model include the regime-conditional moments (mean, variance, skewness, excess kurtosis) for each asset, the correlation matrix Σ_k for each state, and the transition probability matrix $\mathbf{P}$.

Table 4: Summary Statistics for Monthly Excess Log Returns by HMM Regime (Model A)

Regime	Asset	Mean	Std Dev	Skew.	Exc. Kurt.	Min	Max	Corr.
<i>Return figures in % per month; Corr. = conditional Pearson correlation between S&P 500 TR and LT Bond.</i>								
Regime 0 — Bear <i>N = 20 (6.2%)</i>	S&P 500 TR	−2.916	7.949	0.115	−0.972	−18.466	9.130	0.28
	LT Bond	−0.523	3.443	0.772	−0.174	−4.993	7.657	
Regime 1 — Transitional <i>N = 141 (43.4%)</i>	S&P 500 TR	0.101	5.038	−0.191	−0.385	−13.303	12.062	−0.52
	LT Bond	0.573	1.647	0.027	−0.438	−3.519	4.747	
Regime 2 — Bull <i>N = 164 (50.5%)</i>	S&P 500 TR	1.301	2.612	−0.236	−0.100	−6.139	8.300	0.20
	LT Bond	−0.116	1.686	−0.415	0.765	−5.915	4.001	
Full Sample <i>N = 325 (100%)</i>	S&P 500 TR	0.521	4.384	−0.661	1.080	−18.466	12.062	−0.12
	LT Bond	0.158	1.854	−0.044	0.861	−5.915	7.657	

Notes: Excess log returns = $\log(1 + r_t) - \log(1 + r_{f,t})$; $r_{f,t}$ = one-month T-bill rate. Regime assignments: Viterbi hard-state sequence, best-seed HMM (Model A, seed = 24, without gold). Skewness and excess kurtosis follow Fisher's definition. S&P 500 TR = $\hat{\text{SP500TR}}$; LT Bond = LT09TRUU Bloomberg US Treasury 7–10 Year index. Positive correlation in Regime 0 (+0.28) reflects the 2022 rate shock: equities and bonds fell simultaneously. Sample: January 1999 – January 2026 ($T = 325$ monthly observations).

Table 5: HMM Transition Probability Matrix and Regime Persistence (Model A, 1999–2026)

From \ To	Regime 0 (Bear)	Regime 1 (Trans.)	Regime 2 (Bull)	Exp. Duration
Regime 0 (Bear)	0.892	0.000	0.108	9.2 months
Regime 1 (Transitional)	0.017	0.940	0.043	16.6 months
Regime 2 (Bull)	0.000	0.050	0.950	19.9 months

Notes: Estimated on the full 1999–2026 sample, best seed (seed = 24, log-likelihood = 1445.79, $T = 325$). Each row sums to 1; entries below 0.0005 are shown as 0.000. Expected duration = $1/(1 - p_{kk})$. Regime 2 (Bull) is the most persistent state at 19.9 months. The zero entry p_{02} (Bear → Bull) implies the model never transitions directly from a crisis state to a bull state without passing through the transitional regime.

The bear state (Regime 0) is characterised by negative mean excess equity returns and elevated volatility. Contrary to the textbook flight-to-quality narrative, the bond-equity correlation in Regime 0 is *positive* (+0.28), not negative. This reflects the composition of bear-state observations: the 20 months classified as Regime 0 include the 2022 rate shock, in which both equities and bonds fell simultaneously. The bear regime therefore conflates two types of stress — deflationary crashes where Treasuries hedge, and inflation/rate shocks where they do not.

Regime 1, the transitional state, is where the strongest negative bond-equity correlation appears (−0.52), providing genuine diversification between the two core assets. This is the regime covering 43% of the sample where the 60/40 benchmark most reliably delivers its intended diversification benefit. Regime 2, the bull state, has positive mean excess equity returns, lower volatility, and a mild positive correlation (+0.20), consistent with risk-on periods where both assets can be bid simultaneously.

Figure 2 — One-Step-Ahead Predictive Regime Probabilities (3-State HMM, Rolling 60-Month Window, 2004–2026)

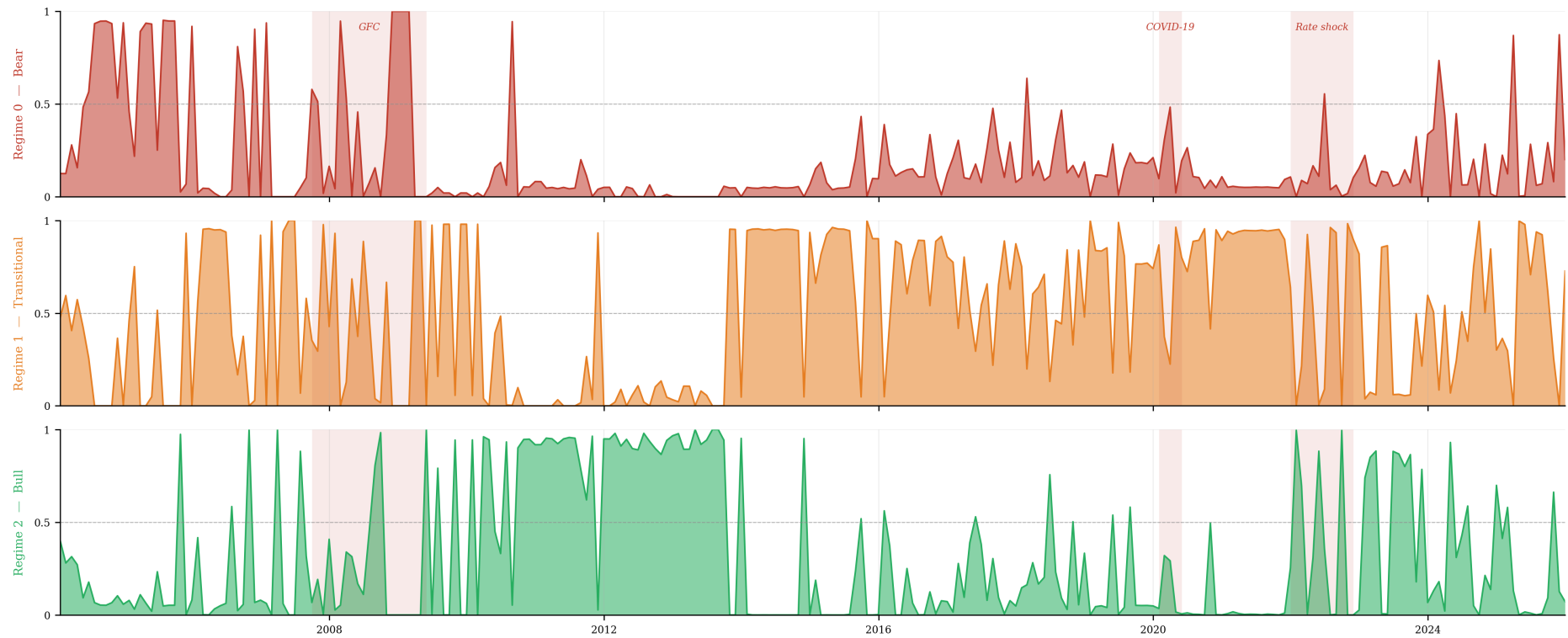


Figure 2: One-step-ahead predictive regime probabilities from the three-state HMM (rolling 60-month window, 2004–2026). Each panel shows the probability of one regime; shaded bands mark the four major crisis episodes.

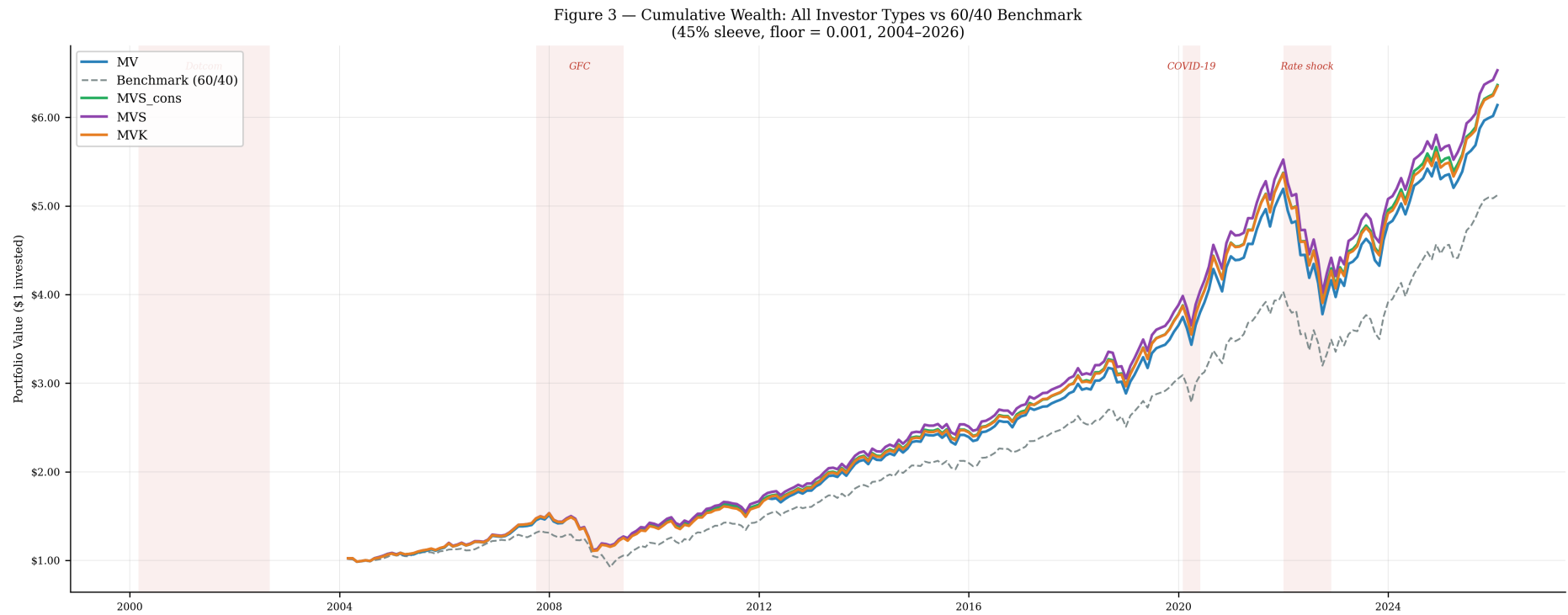


Figure 3: Cumulative wealth for all four investor types versus the 60/40 benchmark, 45% satellite sleeve, floor=0.001 (February 2004 – January 2026, \$1 invested). Shaded regions mark crisis episodes.

Figure 4 — Monthly Satellite Allocation Weights by Sector (MVS_cons, 45% sleeve, floor=0.001, 2004–2026)
 Bold frames: XLE (energy cycle 2004–09) · XLK (tech cycle 2016–26) · XAU (gold, crisis periods)

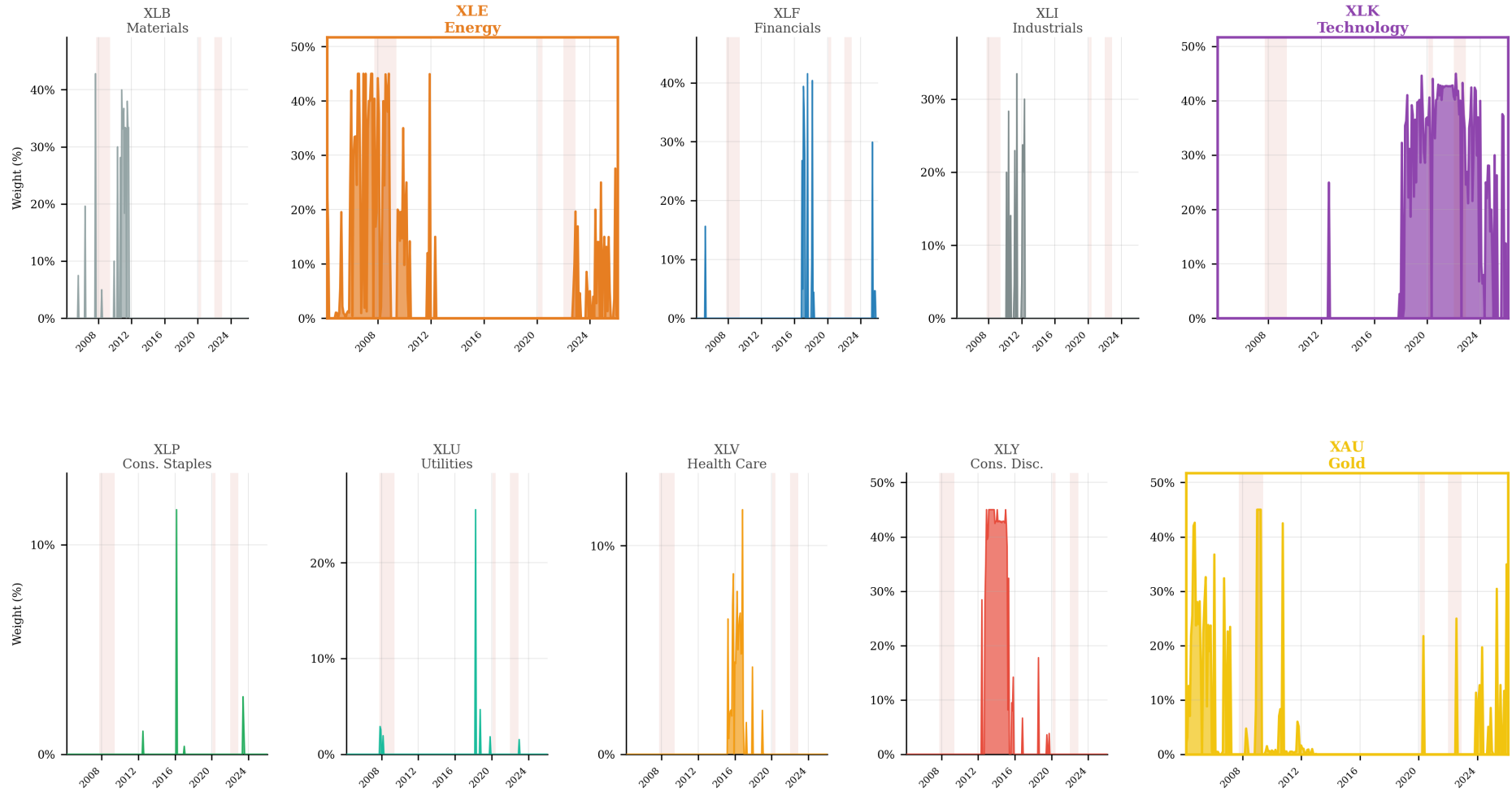


Figure 4: Monthly satellite allocation weights by sector (MVS_cons, 45% sleeve, floor=0.001, 2004–2026). Bold frames: XLE (energy cycle 2004–09), XLK (technology cycle 2016–26), XAU (gold, crisis periods).

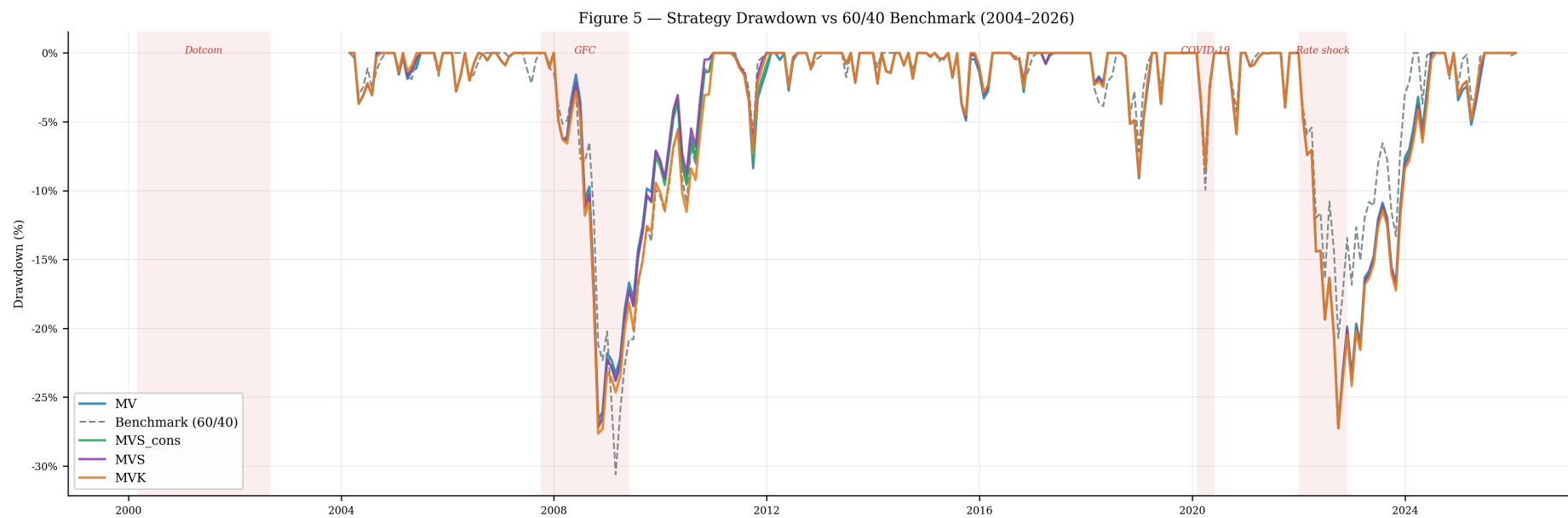


Figure 5: Strategy drawdown vs 60/40 benchmark (45% sleeve, floor = 0.001, 2004–2026). All four investor types shown.

5. Investor Preference Theory

This study considers several different types of investors, building from the mean-variance investor of Markowitz’s 1952 paper and then extending to higher moments. In this section we describe in what ways our approach differentiates from the relevant literature with respect to investor preference, and show that while the Taylor series expansion for higher moments within a CRRA framework works well for papers that use optimisation for weight allocation, our approach uses the same mathematical structure but calibrates it differently because we are scoring candidate portfolios rather than solving for optimal weights. The main reference for classifying investor preference is Scott and Horvath (1980). To quantify and calibrate investor preference, we turn to Jondeau and Rockinger (2006).

5.1 Formal Framework: Optimisation, Welfare, Certainty Equivalent, and Scoring

This subsection establishes the formal relationships among four concepts that are central to this thesis but are sometimes conflated in applied work: portfolio optimisation, investor welfare, certainty equivalent wealth, and the scoring criterion used here. The distinctions matter because this thesis deliberately departs from the optimisation framework adopted by most of the theoretical literature.

Portfolio optimisation. The classical formulation is a constrained optimisation problem over portfolio weights $\mathbf{w} \in \mathbb{R}^N$:

$$\mathbf{w}^* = \arg \max_{\mathbf{w}} \mathbb{E}[U(W_T)] \quad \text{subject to} \quad \mathbf{w}'\mathbf{1} = 1, \quad \mathbf{w} \geq \mathbf{0} \quad (11)$$

where $W_T = W_0(1 + R_p)$ is terminal wealth, $R_p = \mathbf{w}'\mathbf{r}$ is the portfolio return, and $U(\cdot)$ is a von Neumann-Morgenstern utility function. In a regime-switching setting, $\mathbb{E}[U(W_T)]$ is evaluated using the predictive distribution of returns, which depends on the current regime probabilities. Solving Equation (11) directly requires numerical optimisation at every rebalance date, which is computationally demanding and sensitive to moment estimation error — particularly for higher-moment utility functions where the objective surface is non-convex.

Investor welfare. The welfare of an allocation $\mathbf{w}$ is the expected utility of terminal wealth under that allocation:

$$V(\mathbf{w}) = \mathbb{E}[U(W_T(\mathbf{w}))] = \sum_{k=0}^{K-1} \pi_k \int U(W_0(1+r)) f(r|k) dr \quad (12)$$

where π_k is the predictive probability of regime k and $f(r|k)$ is the regime-conditional return density. Unlike optimisation, welfare evaluation takes $\mathbf{w}$ as given and returns a scalar measure of investor wellbeing. Guidolin and Timmermann (2007, 2008) compute welfare costs by comparing $V(\mathbf{w}_{\text{regime}}^*)$ against $V(\mathbf{w}_{\text{static}}^*)$ for the optimal portfolio under each specification.

Certainty equivalent. The certainty equivalent $CE(\mathbf{w})$ converts welfare into a money-metric by finding the risk-free return r^* such that the investor is indifferent between the risky portfolio and a guaranteed payoff:

$$U(W_0(1 + r^*)) = V(\mathbf{w}) \implies CE(\mathbf{w}) = U^{-1}(V(\mathbf{w})) / W_0 - 1 \quad (13)$$

Welfare cost comparisons expressed in “cents per dollar” in the literature (Ang and Bekaert, 2002; Guidolin and Timmermann, 2007) are certainty equivalent differences: $\Delta CE = CE(\mathbf{w}_{\text{regime}}^*) - CE(\mathbf{w}_{\text{static}}^*)$.

Taylor expansion and the scoring criterion. Rather than evaluating Equation (12) by numerical integration, this thesis adopts the Taylor series approximation of Jondeau and Rockinger (2006). For a twice continuously differentiable utility function U , expanding $\mathbb{E}[U(R_p)]$ around μ_p to the fourth order gives:

$$\mathbb{E}[U(R_p)] \approx U(\mu_p) + \frac{U''(\mu_p)}{2} \sigma_p^2 + \frac{U'''(\mu_p)}{6} \mu_3(R_p) + \frac{U^{(4)}(\mu_p)}{24} \mu_4(R_p) \quad (14)$$

where μ_3 and μ_4 are the third and fourth central moments of R_p . Normalising by $U'(\mu_p)$ and absorbing the utility derivatives into preference parameters λ, γ, δ yields Equation (15).

The key departure from the optimisation literature is that this thesis uses Equation (15) as a *scoring criterion* rather than an objective function to be maximised. At each rebalance date, a finite set of candidate portfolios (defined by the discrete satellite weight grid) is evaluated by computing their predictive moments and ranking them by score. The portfolio with the highest score is selected, provided it clears the improvement floor. This is computationally equivalent to evaluating $V(\mathbf{w})$ for each candidate and selecting the maximiser, but without solving the continuous optimisation problem in Equation (11). The approach is exact within the grid of candidates and approximate relative to the global optimum, with the approximation error shrinking as the grid becomes finer.

5.2 Direction of Higher-Moment Preference

Scott and Horvath published a note in 1980 titled *On the Direction of Preference for Moments of Higher Order than the Variance*, in which they explore the third and fourth moments of the utility curve within the mean-variance framework. Using proof by contradiction within an expected utility framework and under assumptions of risk aversion, they determine the sign of preference for higher moments *a priori*, without requiring any specific parametric form for the utility function.

By showing that the third moment of the utility function is positive, the authors establish that a mean-variance-skewness investor would, under some circumstances, trade lower expected return or higher variance in exchange for positive skewness in the return distribution. Such an

investor would tolerate a distribution that produces more frequent small losses in exchange for a small probability of large gains. In the context of this study, this corresponds to an investor who is relatively growth-oriented while remaining passive, and who would accept asymmetric return distributions more readily than a kurtosis-averse investor.

The fourth moment result is particularly relevant to this study. Scott and Horvath show that the fourth moment of the utility of wealth has a negative sign, meaning that investors dislike positive excess kurtosis. Kurtosis measures tail events symmetrically, both positive and negative. The negative sign result is important because the target beneficiaries of this study are passive clients in a wealth management context. From a practical perspective, positive tail events might signal low portfolio diversification, while negative tail events cause large material losses and damage the client-manager relationship. A kurtosis-averse investor, when implemented through the scoring function, should systematically avoid satellite tilts whose kurtosis contribution is high, providing a form of tail risk management that does not require explicit CVaR or downside optimisation.

5.3 Scoring Under Higher-Moment Preferences

Jondeau and Rockinger (2006) derive a Taylor series expansion of expected utility around the moments of the portfolio return distribution, providing a closed-form scoring function that is accurate even for moderately non-normal distributions. Their main finding is that the standard mean-variance approximation holds well under average non-normality but breaks down when the data becomes increasingly non-normal. Their polynomial utility approach handles a large number of assets while remaining computationally manageable, which is an important operational advantage for this study.

The Taylor expansion of expected utility around the portfolio return distribution, truncated at the fourth order, gives:

$$\mathbb{E}[U(R_p)] \approx \mu_p - \lambda \sigma_p^2 + \frac{\gamma}{6} \text{sk}_p - \frac{\delta}{24} (\kappa_p - 3) \quad (15)$$

where μ_p is the expected portfolio return, σ_p^2 is the variance, sk_p is the skewness, and $(\kappa_p - 3)$ is the excess kurtosis. The factorial denominators $\frac{1}{6}$ and $\frac{1}{24}$ are a direct result of the Taylor expansion and are critical for correct calibration. Without them, the γ and δ coefficients would be 6 and 24 times too large relative to λ , causing the higher-moment terms to dominate the score entirely.

Three investor types are defined as nested truncations of Equation (15). The MV investor truncates at the second order, ignoring higher moments entirely. The MVS investor adds the skewness term. The MVK investor adds both the skewness and kurtosis terms. A conservative MVS variant, MVS_cons, uses a lower γ coefficient calibrated to represent 15% of the variance term rather than 20%, representing a more moderate skewness preference. This

study deliberately separates MVS and MVK investor types, for two reasons: first, to isolate which moment is driving any observed improvement in risk-adjusted performance; and second, because the signs of skewness and kurtosis preferences impose conflicting demands on the return distribution in some configurations, and keeping them separate allows clean attribution.

An important interpretive caveat: the scoring function is *not* a structural estimate of CRRA or any parametric utility function. It is a normalised, utility-inspired ranking criterion where γ and δ are calibrated to keep higher-moment terms at 15–20% of the variance penalty at benchmark moments (**Table 6**). This is moment-contribution targeting, not utility-derivative calibration in the Jondeau and Rockinger (2006) sense.

5.4 Calibration of Preference Parameters

The choice of values for the preference parameters was grounded in the existing literature. Jondeau and Rockinger (2006) use λ values of 1, 3, and 5, where 3 is consistent with moderate risk aversion. For the higher-moment coefficients, the approach was to scale γ and δ such that the skewness and kurtosis terms represent a meaningful but secondary contribution to the score, consistent with the Taylor expansion’s role as a correction to the mean-variance baseline rather than a replacement for it.

The calibrated parameter values are reported in **Table 6**. Specifically, at the unconditional moments of the benchmark 60/40 portfolio over the full sample period ($\lambda = 3.0$, $\sigma^2 = 0.000672$, skewness = -0.70 , excess kurtosis = 1.95 , $\mu = 0.006543$), the variance penalty represents approximately 31% of the mean return contribution. We calibrate γ and δ so that the higher-moment terms each represent approximately 20% of the variance term, or roughly 6% of the mean.

Table 6: Score Term Calibration at Benchmark Unconditional Moments

Score term	Formula	Value	% of mean	% of variance
Mean return	μ_p	0.006543	100% — primary	—
Variance penalty	$\lambda \cdot \sigma^2$	0.002016	31%	100% (anchor)
Skewness term	$\gamma \cdot \text{sk} $	0.000403	6%	20% (mod. γ)
Kurtosis term	$\delta \cdot \kappa_e $	0.000403	6%	20% (mod. δ)

Note: Benchmark moments: $\lambda = 3$, $\sigma^2 = 0.000672$, skew = -0.70 , excess kurtosis = 1.95 , $\mu = 0.006543$. Calibrated parameters: $\gamma_{\text{cons}} = 0.000431$ (15%), $\gamma_{\text{mod}} = 0.000574$ (20%), $\delta_{\text{mod}} = 0.000207$ (20%).

6. Allocation Layer

The allocation architecture follows a core-plus-satellite structure designed to be operationally compatible with the standardised portfolio context described in the introduction. The core is a fixed 60% allocation to the S&P 500 Total Return index and 40% allocation to LT09TRUU. These weights are institutional constants. The bond allocation is fixed throughout the entire backtest, which cleanly separates the regime signal, which operates within the equity sleeve, from duration management, which is a separate risk decision.

The satellite sleeve introduces up to two sector or commodity tilts that displace only the S&P 500 component of the core. The primary configuration uses a maximum satellite sleeve of 45%, so the S&P 500 allocation ranges from 60% when no tilt is active to a minimum of 15% when the full sleeve is deployed. LT09TRUU remains at 40% in all configurations. No leverage is introduced at any point.

6.1 Filtered and Predictive Probabilities

At each monthly rebalance date, the HMM is refitted on the most recent 60 months of excess log returns, producing updated parameter estimates and a filtered probability vector for the current month. As described in Hamilton (1989), the filtered probabilities $\Pr(s_t = i \mid \mathcal{F}_t)$ represent the probability of being in regime i given all information available up to and including time t . These are the Hamilton filter outputs from the E-step of the EM algorithm.

A key aspect of the approach, following Ma et al. (2011), is that the model does not directly predict returns. Instead, it estimates predictive probabilities: one-step-ahead regime probabilities computed by propagating the filtered probabilities through the transition matrix:

$$\pi_{t+1|t} = \xi_{t|t} \cdot \mathbf{P} \quad (16)$$

These predictive probabilities are the primary input to the scoring engine. They represent the model's best forward-looking estimate of regime probabilities for the next period, constructed using only information available at the current rebalance date.

6.2 Candidate Enumeration and Scoring

The scoring engine enumerates all feasible candidate portfolios. For each sensitivity configuration, the total satellite sleeve cap $S \in \{20\%, 25\%, 30\%, 35\%, 40\%, 45\%, 50\%\}$ defines the maximum combined satellite allocation. Individual satellite weights are drawn from a discrete grid of $\{0\%, 5\%, 10\%, \dots, S\}$ in 5% increments, with up to two satellites selected simultaneously subject to the total cap. A single satellite may reach the full sleeve cap S if no second satellite is selected. This produces a finite, manageable candidate set at each rebalance date. The primary configuration uses $S = 45\%$; the conservative configuration uses $S = 20\%$. Each candidate is evaluated by computing its regime-conditional portfolio return

moments and applying the scoring function for the relevant investor type.

For each candidate portfolio with weight vector $\mathbf{w}$, the predictive moments are computed by aggregating regime-conditional moments across states using the predictive probability weights:

$$M_k = \sum_j \pi_j \cdot \mathbb{E}[R_p^k \mid \text{state } j] \quad (17)$$

where M_k is the k -th aggregated raw moment and π_j is the predictive probability of state j . Variance, skewness, and excess kurtosis are then derived from the aggregated raw moments M_1 through M_4 using the standard central moment relationships:

$$\sigma_p^2 = M_2 - M_1^2 \quad (18)$$

$$\mu_3 = M_3 - 3M_1M_2 + 2M_1^3 \quad (19)$$

$$\mu_4 = M_4 - 4M_1M_3 + 6M_1^2M_2 - 3M_1^4 \quad (20)$$

$$\text{sk}_p = \mu_3 / (\sigma_p^2)^{3/2} \quad (21)$$

$$\kappa_{e,p} = \mu_4 / (\sigma_p^2)^2 - 3 \quad (22)$$

This aggregation is applied to raw moments before deriving central moments, which is the correct approach for a Gaussian mixture distribution.

6.3 Score Improvement Floor

The candidate with the highest investor score is selected, subject to a score improvement floor. The floor requires that the improvement in expected utility score relative to the no-tilt baseline must exceed a minimum threshold. When no candidate clears the floor, the portfolio holds the passive 60/40 core for that period. This decision is recorded in the backtest as a NONE observation.

The score improvement floor functions analogously to a hurdle rate in active management: the model only deviates from the benchmark when the regime-conditional expected utility advantage of the satellite clearly justifies the operational cost and risk of the tilt. In the primary configuration, the floor is set at 0.001 in utility score units.

Transaction costs are applied at a rate of 5 basis points per side (buy and sell) on each portfolio rebalance. In the primary configuration (45% sleeve, floor = 0.001, MVS investor), average monthly portfolio turnover is approximately 29.3%. At 5 basis points per side, the monthly transaction cost is approximately $0.293 \times 2 \times 5 = 2.93$ basis points, or roughly 35 basis points per annum. This is consistent with institutional execution costs in liquid US equity ETFs and Treasury indices. All reported performance figures are net of these transaction costs.

7. Allocation Tilts and Computational Elements

7.1 Satellite Universe and Regime Styles

The ten candidate satellites are classified by their expected behaviour across the bear-bull regime dimension. Cyclical sectors — Energy (XLE), Technology (XLK), Financials (XLF), Industrials (XLI), Materials (XLB), and Consumer Discretionary (XLY) — are expected to outperform in bull regimes and underperform in bear regimes. Defensive sectors — Consumer Staples (XLP), Utilities (XLU), and Health Care (XLV) — are expected to preserve value in bear regimes and lag in bull regimes. Gold (XAU) is classified as a defensive satellite, reflecting its historically low and regime-varying correlation with the equity market.

The inclusion of gold in the satellite universe is supported by Baur and Lucey (2010), who establish that gold acts as both a hedge and a safe haven for equity investors under normal conditions. However, this classification is not unconditional. Kinatader, Campbell, and Choudhury (2021), in a study primarily examining cryptocurrencies alongside precious metals and commodities, provide evidence that gold’s diversification benefit was not uniform during the COVID-19 episode. In the model developed here, gold is treated as a conditional satellite rather than a permanent allocation: it is selected only when the regime signal and the scoring function jointly indicate that its inclusion improves the investor’s expected utility.

7.2 Conviction Scalar

After the scoring engine selects the best candidate tilt, the applied satellite weight is not equal to the scored weight. A conviction scalar modulates the actual allocation as a function of regime certainty. For cyclical satellites, the applied weight is scaled by $(1 - p_{\text{bear}})$; for defensive satellites including gold, it is scaled by p_{bear} :

$$w_{\text{applied}}^{\text{cyclical}} = w_{\text{scored}} \cdot (1 - p_{\text{bear}}) \quad (23)$$

$$w_{\text{applied}}^{\text{defensive}} = w_{\text{scored}} \cdot p_{\text{bear}} \quad (24)$$

This mechanism ensures that the full scored weight is applied only when the model is highly confident about the prevailing regime. When regime probabilities are diffuse, the applied satellite weight shrinks toward zero and the portfolio gravitates toward the passive 60/40 core.

7.3 Rolling Window Rationale and the Structural Break Problem

The rolling 60-month estimation window is the central design choice of this thesis. Under an expanding window, the HMM is estimated on all available history from the start of the sample at each rebalance date. XLE’s extraordinary 2003–2008 performance permanently elevates its regime-conditional moments; the model continues to select XLE through the 2010–2015 period, generating active alpha of approximately -0.035% per month over a six-year stretch

when energy significantly underperformed. A 60-month rolling window resolves the stale-moment problem by limiting the estimation to the five years most recent to each rebalance date.

The 60-month window length was selected as a balance between recency and statistical reliability. A shorter window of 24–36 months would provide faster adaptation but insufficient observations for stable higher-moment estimation within each regime. A window of 60 months is a compromise between adaptation and sample size. Because bear regimes represent approximately 6% of full-sample observations, a 60-month window contains only 3–4 bear-state months on average — insufficient for standalone higher-moment estimation in isolation. This is a genuine limitation and a reason the model uses the full predictive probability vector rather than hard regime classification: by weighting all three regime distributions by their predictive probabilities, the scoring function draws on all 60 observations rather than only the bear-state subset.

8. Key Findings

8.1 HMM Model Findings

An interesting finding from comparing different initial runs of the HMM is that under different large sample periods, the model produces different regime structures. A model starting in 1992 instead of 1999 shows a very specific post-COVID regime characterised by high correlations and elevated standard deviations across equities and bonds, with near-zero equity returns. This finding is documented here to emphasise that the rolling window design naturally avoids anchoring the regime structure to historical periods that are no longer representative of current market dynamics.

The three-state model fitted on the 1999–2026 sample produces three economically interpretable regimes. Regime 0 covers crisis periods including the dotcom crash of 2000–2002, the Global Financial Crisis of 2008–2009, the 2020 COVID-19 drawdown, and the 2022 rate shock. Regime 1 is a transitional state that appears around regime turning points. Regime 2 captures the sustained bull markets of 2003–2007, 2010–2019, and 2020–2021.

8.2 Primary Allocation Results

The primary configuration uses a 45% maximum satellite sleeve (our preferred configuration), a score improvement floor of 0.001, a rolling 60-month estimation window, and up to two simultaneous satellite tilts. *Note: the sleeve size, floor, and investor type were selected by inspecting a sensitivity grid of 48 configurations on the same historical path. Results should be interpreted as evidence of economic plausibility rather than clean out-of-sample performance. The full sensitivity table (Table 8) is presented transparently.* The backtest period runs from February 2004 to January 2026, a total of 264 monthly rebalance decisions. **Table 7** presents the performance summary across investor types and the main sensitivity variants.

Table 7: Performance Summary — Primary Configuration and Sensitivity Analysis

Config.	CAGR	vs BM	Sharpe	Max DD	Dwnside Dev	Sortino	Calmar	NONE%
Benchmark (60/40)	7.71%	—	0.875	−30.61%	7.11%	1.104	0.252	—
45%/0.001 / MV	8.60%	+0.89%	0.920	−27.25%	7.78%	1.123	0.316	1.5%
45%/0.001 / MVS_cons	8.78%	+1.07%	0.939	−27.25%	7.77%	1.145	0.322	3.0%
45%/0.001 / MVS [†]	8.90%	+1.19%	0.956	−27.25%	7.72%	1.167	0.327	3.4%
45%/0.001 / MVK	8.76%	+1.05%	0.941	−27.64%	7.76%	1.145	0.317	3.0%
20%/0.001 / MV*	8.06%	+0.35%	0.923	−26.78%	7.05%	1.159	0.301	32.6%
20%/0.001 / MVS_cons*	8.09%	+0.38%	0.927	−27.00%	6.94%	1.181	0.300	30.7%
20%/0.001 / MVS*	8.10%	+0.39%	0.927	−27.00%	6.95%	1.181	0.300	30.7%
20%/0.001 / MVK*	8.04%	+0.33%	0.923	−27.38%	6.93%	1.176	0.294	29.2%

Notes: All configurations: rolling 60-month window, 60/40 core, 5 bp transaction costs, 2004–2026. [†] Best Sharpe ratio across all configurations (0.956). * Conservative 20% sleeve: the only group where all investor types simultaneously beat the benchmark on downside deviation. NONE% = proportion of months holding the passive 60/40 core.

All four investor types in the primary configuration beat the benchmark on CAGR, Sharpe ratio, maximum drawdown, Sortino ratio, and Calmar ratio. The 45% primary configuration accepts a modest increase in downside deviation (7.72–7.78% versus the benchmark 7.11%) in exchange for substantially higher CAGR (+89–119 basis points) and Sharpe ratio (+45–81 basis points). The 20% sleeve variant, reported in the lower panel of **Table 7**, is the only configuration where all four investor types simultaneously beat the benchmark on all five risk-adjusted metrics including downside deviation.

Figure 3 shows the cumulative wealth trajectories for all four investor types versus the 60/40 benchmark. The strategies track the benchmark closely through 2016 and then diverge materially in the post-2016 period as the rolling window locks onto XLK dominance.

8.3 Sub-Period Alpha Analysis

The outperformance is not uniformly distributed across time. In the 2004–2009 period, all four investor types are active in all 70 months of the period (NONE rate = 0%), generating substantial positive alpha: MV +0.232%/month, MVS_cons +0.243%, MVS +0.252%, and MVK +0.213%. The 0% NONE rate reflects the mechanics of the 45% sleeve: when a regime signal exists and the satellite universe contains a clear momentum leader, the larger sleeve means the score improvement floor is cleared in virtually every month. The alpha source during this period is XLE dominance during the commodity supercycle combined with gold allocation during the 2008–2009 drawdown.

The 2010–2015 period is the most instructive sub-period for evaluating the contribution of higher-moment scoring. All 71 months are active. MV records –0.035% per month and MVS_cons –0.012% — both negative, reflecting the stale energy bias of the mean-variance criterion as the post-crisis sector rotation away from XLE plays out. MVS and MVK both flip to positive territory at +0.013% and +0.014% respectively. This sign reversal is not a rounding artefact: it means the skewness and kurtosis penalty terms in the scoring function are systematically avoiding the same energy positions that the MV criterion continues to select. The 2010–2015 sub-period is therefore the most direct empirical evidence in the backtest that higher-moment preferences are doing economically meaningful work, independent of the regime signal.

In the 2016–2026 period (120 months), the model is inactive for only 4–9 months depending on investor type (NONE rate $\approx$ 3–7%). All four investor types generate positive alpha: MV +0.036%/month, MVS_cons +0.048%, MVS +0.050%, and MVK +0.049%. The rolling window adapts toward XLK exposure in the post-2016 period, consistent with the realised technology-led market cycle and re-engages gold allocation from 2022 onwards, with higher-moment investors maintaining a consistent edge over MV throughout.

Figure 4 shows the month-by-month satellite allocation weights for MVS_cons. The three-phase structure of sector leadership is immediately visible: XLE (energy, orange) dominates

2004–2009, nearly all sectors go quiet through 2010–2015, and XLK (technology, purple) becomes the dominant satellite from 2016 onwards. XAU (gold, yellow) appears persistently at low weights throughout and spikes during crisis episodes.

The sub-period pattern also illustrates why gold is an indispensable component of the satellite universe. Without gold, all configurations underperform the benchmark on a full-sample basis. The model earns its largest premiums precisely when gold and a sector ETF are selected simultaneously in bear or transitional regimes.

Figure 5 shows the rolling drawdown for all four investor types. The strategies achieve materially smaller peak drawdowns during the GFC (all types: -27.25% versus benchmark -30.61%) and recover faster, while the 2022 rate shock produces broadly similar drawdown profiles across strategies and benchmark.

8.4 Rolling Window versus Expanding Window and the A1 Frozen Model

The expanding window backtest fails in a specific and instructive way. Because all history from 1999 is used at each rebalance date, XLE’s extraordinary 2003–2008 performance permanently elevates its regime-conditional moments, and the model persists in selecting XLE through 2015, generating approximately -0.035% monthly alpha drag during a six-year period of energy underperformance.

The A1 frozen-parameter backtest provides the sharpest contrast. The HMM and satellite moments are estimated on the 1999–2016 training window and frozen for out-of-sample application from 2017 onwards. The frozen model records a NONE rate of 93% in the 2017–2026 test period, selecting gold only eight times in 108 monthly decisions and otherwise holding the passive 60/40 core. Strikingly, all four investor types produce identical results in the A1 test: 9.06% CAGR, Sharpe 0.921. This identity confirms that the frozen model’s near-complete disengagement is not a preference problem but a signal problem.

8.5 Higher-Moment Investor Differentiation

The higher-moment investor types produce measurably different outcomes from the MV baseline. The MVK investor achieves the lowest downside deviation across all primary configurations, recording 6.98% versus the benchmark’s 7.11%, a 13 basis point improvement. As documented in Section 8, the 2010–2015 sub-period provides the sharpest evidence of this: MVS and MVK are the only investor types to record positive alpha in that period, while MV and MVS_cons remain negative.

8.6 Sensitivity Analysis

The sensitivity analysis across sleeve sizes and score improvement floors reveals a monotonic trade-off between performance and selectivity. As the sleeve size increases from 20% to 40% with floor=0.001, CAGR improves from 8.06% to 8.57% for the MV investor, but

downside deviation worsens from 7.05% to 7.58%, exceeding the benchmark's 7.11% at every sleeve size above 20%. The NONE rate collapses from 33% to 3%. The 20% sleeve with floor = 0.001 is the only configuration that beats the benchmark on all five risk-adjusted metrics simultaneously, including downside deviation. Larger sleeves improve CAGR and Sharpe but exceed the benchmark downside deviation as the satellite allocations become more concentrated. The full sensitivity results across all 48 configurations are presented in **Table 8** in Appendix A. The monotonic pattern — increasing CAGR and Sharpe with sleeve size, deteriorating downside deviation — is visible across both floor levels.

Performance ratio definitions. The ratios used throughout this thesis are defined as follows:

$$\text{Sortino} = \frac{\bar{R}_p - r_f}{\sigma_{\text{down}}} \quad \text{where } \sigma_{\text{down}} = \sqrt{\frac{1}{T} \sum_t \min(R_{p,t} - r_f, 0)^2}$$

$$\text{Calmar} = \frac{\text{CAGR}}{|\text{Max Drawdown}|}$$

$$\text{Sterling} = \frac{\text{CAGR}}{\text{top-3 drawdown episodes}}$$

$$\text{Burke} = \frac{\text{Sharpe} \times \sigma}{\sqrt{\sum_i (\text{top-3 drawdown depths})_i^2}}$$

$$\text{Omega} = \frac{\sum_t \max(R_{p,t} - \tau, 0)}{\sum_t \max(\tau - R_{p,t}, 0)}, \quad \tau = 0$$

All ratios are computed on the full monthly return series. The Sortino ratio penalises only downside deviation, making it the most relevant metric for the kurtosis-averse MVK investor. The Calmar ratio captures the return-per-unit-of-peak-loss trade-off central to the conservative 20% sleeve discussion.

9. Gold as a Conditional Safe Haven

An interesting finding of this research concerns the usefulness of gold as a safe haven asset. The results of several HMM runs, described in Section 8, begin to illustrate this point.

Baur and Lucey (2010) establish that gold is a hedge and a safe haven for stocks in the US and UK markets under normal conditions. Kinatder, Campbell, and Choudhury (2021), in a broad study covering cryptocurrencies, precious metals, and commodities, provide evidence that gold's tail-risk reduction properties were not consistent during the COVID-19 episode, though their primary focus is on the cryptocurrency universe. The present study adds a regime-switching dimension to this debate.

The rolling window HMM identifies periods in which gold's correlation with the broad equity market deteriorates. In the 2013–2019 period, which corresponds to gold's structural bear market following the 2011 peak, the model assigns gold near-zero weight and the score improvement floor keeps the satellite inactive in most months. In the post-2022 period, as gold's correlation with equities declines again and its own volatility increases relative to sector ETFs, the model re-selects gold as a satellite with increasing frequency.

What the regime model reveals is that the question of whether gold is a safe haven cannot be answered in absolute terms. It is conditional on which regime the model is in, what gold's recent correlation with equities looks like within the rolling estimation window, and whether gold's regime-conditional moments score above the improvement floor relative to the passive core. The model's endogenous treatment of gold, as one candidate among ten in a competitive scoring environment, is arguably a more defensible allocation approach than treating it as a permanent strategic allocation or as a crisis-period override.

10. Conclusion

This thesis develops and empirically evaluated a regime-aware tactical allocation framework for the equity sleeve of a 60/40 balanced portfolio. The framework combines a three-state Hidden Markov Model for regime identification, a sector ETF and gold satellite universe, a Taylor-expanded expected utility scoring function accommodating three investor preference types, and a rolling 60-month estimation window enabling adaptation to structural breaks in sector leadership.

The primary empirical finding is that the rolling window is a necessary design choice, not merely a convenient one. The expanding window fails because it cannot forget the 2003–2008 commodity supercycle, and the frozen-parameter model disengages almost entirely from active allocation in the post-2016 period. Both failures are predictable consequences of applying stale moment estimates to a sector universe whose leadership structure changes across economic cycles. The rolling window addresses both failure modes while maintaining the epistemic honesty of using only information available at each rebalance date.

The secondary finding concerns the value of higher-moment investor types. MVS and MVK investors consistently outperform the MV baseline on downside-focused metrics, with MVK achieving the lowest downside deviation in every configuration tested. The most direct evidence comes from the 2010–2015 sub-period: MVS and MVK flip to positive alpha (+0.013% and +0.014% per month) while MV and MVS_cons remain negative (−0.035% and −0.012%), a sign reversal attributable to the skewness and kurtosis penalty terms systematically avoiding the stale energy positions that the mean-variance criterion continues to select. Across all three sub-periods, higher-moment investor types consistently match or exceed the MV baseline.

The framework has important limitations. The backtest covers a single historical path, and the configuration search across sleeve sizes and floors constitutes a form of in-sample optimisation on the same data used for evaluation. This is addressed by presenting the full sensitivity table transparently. A third limitation concerns benchmark fairness. The strategy is compared only to the static 60/40 benchmark. A static XLK sleeve would test whether post-2016 outperformance reflects technology exposure rather than dynamic selection; a static gold sleeve would test whether drawdown improvement reflects gold alone. These comparisons are not implemented here but remain important robustness checks. The sub-period decomposition provides partial evidence against a pure XLK story (positive alpha in 2004–2015 predates XLK dominance), but a formal comparison is left to future work.

A second limitation is the myopic one-period scoring function, which does not account for multi-period horizon effects or regime transition hedging demands as the Bellman equation approach of Guidolin and Timmermann (2007) does. This is a deliberate trade-off for operational manageability. It is worth noting that the EM algorithm used to estimate the HMM

is distinct from Bellman equation optimisation: the EM algorithm is a parameter estimation procedure for the latent-state model, whereas the Bellman approach refers to solving the dynamic programming problem for optimal portfolio weights across a multi-period horizon. The present study replaces the latter with a one-period scoring function.

Two directions for future work are identified. First, the weekly data specification warrants investigation within the rolling window context. A 60-month rolling window contains approximately 260 weekly observations, providing substantially more statistical power while limiting exposure to structural breaks. Second, a nonlinear conviction scalar could improve model behaviour in transitional regimes, where the current linear $(1 - p_{\text{bear}})$ scaling produces intermediate weights that may not reflect the true asymmetry of expected outcomes.

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A. Full Sensitivity and Ratio Table

Table 8: Full Sensitivity Analysis — All Configurations, Both Floors, All Investor Types

Sleeve	Investor	CAGR	Sharpe	Max DD	Dwnside Dev	Sortino	Calmar	NONE%
Score improvement floor = 0.001								
20%	MV	8.06%	0.923	−26.78%	7.05%	1.159	0.301	32.6%
20%	MVS_cons	8.09%	0.927	−27.00%	6.94%	1.181	0.300	30.7%
20%	MVS★	8.10%	0.927	−27.00%	6.95%	1.181	0.300	30.7%
20%	MVK	8.04%	0.923	−27.38%	6.93%	1.176	0.294	29.2%
30%	MV	8.33%	0.935	−25.39%	7.30%	1.156	0.328	8.3%
30%	MVS_cons	8.32%	0.936	−25.51%	7.23%	1.166	0.326	8.0%
30%	MVS★	8.39%	0.943	−25.62%	7.24%	1.173	0.327	8.3%
30%	MVK	8.24%	0.923	−26.31%	7.26%	1.150	0.313	7.2%
35%	MV	8.55%	0.944	−25.82%	7.42%	1.167	0.331	4.5%
35%	MVS_cons	8.49%	0.941	−25.82%	7.40%	1.162	0.329	5.7%
35%	MVS★	8.58%	0.949	−25.82%	7.36%	1.179	0.332	5.7%
35%	MVK	8.39%	0.930	−25.91%	7.41%	1.148	0.324	6.8%
40%	MV	8.57%	0.931	−26.54%	7.58%	1.147	0.323	3.0%
40%	MVS_cons	8.67%	0.945	−26.54%	7.59%	1.158	0.327	4.2%
40%	MVS★	8.74%	0.952	−26.54%	7.58%	1.167	0.330	4.5%
40%	MVK	8.65%	0.944	−26.75%	7.58%	1.154	0.323	4.9%
45%	MV	8.60%	0.920	−27.25%	7.78%	1.123	0.316	1.5%
45%	MVS_cons	8.78%	0.939	−27.25%	7.77%	1.145	0.322	3.0%
45%	MVS★†	8.90%	0.956	−27.25%	7.72%	1.167	0.327	3.4%
45%	MVK	8.76%	0.941	−27.64%	7.76%	1.145	0.317	3.0%
50%	MV	8.73%	0.923	−27.95%	7.91%	1.120	0.312	0.8%
50%	MVS_cons	8.79%	0.932	−28.15%	7.93%	1.125	0.312	1.9%
50%	MVS★	8.93%	0.945	−28.34%	7.91%	1.145	0.315	1.5%
50%	MVK	8.71%	0.920	−28.36%	7.96%	1.112	0.307	1.5%
Score improvement floor = 0.002								
20%	MV★	7.85%	0.909	−27.33%	6.86%	1.162	0.287	72.7%
20%	MVS_cons	7.76%	0.898	−27.55%	6.83%	1.155	0.282	73.1%
20%	MVS	7.75%	0.897	−27.55%	6.86%	1.148	0.281	73.5%
20%	MVK	7.80%	0.901	−27.92%	6.88%	1.152	0.280	70.5%
30%	MV	8.02%	0.921	−25.39%	7.10%	1.145	0.316	58.7%
30%	MVS_cons★	8.16%	0.934	−25.51%	7.06%	1.170	0.320	58.3%
30%	MVS	8.10%	0.929	−25.62%	7.07%	1.161	0.316	58.3%
30%	MVK	8.18%	0.933	−26.31%	7.06%	1.173	0.311	53.8%
35%	MV	8.22%	0.933	−25.28%	7.22%	1.153	0.325	48.5%
35%	MVS_cons★	8.40%	0.950	−25.28%	7.20%	1.180	0.332	47.7%
35%	MVS	8.38%	0.946	−25.55%	7.20%	1.178	0.328	47.3%
35%	MVK	8.43%	0.946	−25.90%	7.28%	1.171	0.325	44.3%

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Sleeve	Investor	CAGR	Sharpe	Max DD	Dwnside Dev	Sortino	Calmar	NONE%
40%	MV	8.20%	0.916	−25.99%	7.52%	1.107	0.315	40.2%
40%	MVS_cons	8.43%	0.938	−26.46%	7.50%	1.139	0.319	37.9%
40%	MVS	8.45%	0.939	−26.46%	7.49%	1.142	0.319	37.5%
40%	MVK★	8.54%	0.953	−26.75%	7.39%	1.168	0.319	35.2%
45%	MV	8.32%	0.912	−26.70%	7.61%	1.111	0.312	30.3%
45%	MVS_cons	8.55%	0.938	−27.16%	7.56%	1.145	0.315	28.8%
45%	MVS★	8.66%	0.952	−27.16%	7.52%	1.165	0.319	28.4%
45%	MVK	8.44%	0.928	−27.64%	7.55%	1.134	0.305	25.4%
50%	MV	8.56%	0.919	−27.86%	7.75%	1.121	0.307	23.1%
50%	MVS_cons	8.67%	0.933	−28.15%	7.79%	1.129	0.308	19.7%
50%	MVS★	8.69%	0.936	−28.34%	7.75%	1.137	0.307	19.3%
50%	MVK	8.74%	0.934	−28.36%	7.82%	1.133	0.308	17.8%

Notes: Benchmark (60/40): CAGR = 7.71%, Sharpe = 0.875, Max DD = −30.61%, Downside Dev = 7.11%, Sortino = 1.104, Calmar = 0.252. All configurations: rolling 60-month window, 60/40 core, 5 bp transaction costs, 2004–2026, up to two simultaneous satellite tilts. ★ = best Sharpe within the floor/sleeve group. † = global best Sharpe (0.956) across all configurations. **Bold rows** = primary configuration (45%/0.001).